What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Renin–Angiotensin–Aldosterone System Inhibitors

Last updated January 30, 2025 · 13 min read · ScholarRx

Adapted by: Trager Hintze, Diptiman Bose

Editors: Jordan Finn, Russell Johnson

Adapted from: Renin-Angiotensin-Aldosterone System Inhibitors

Listen to this Brick

0:00 / 23:38 1x

Learning Objectives (5) Versions

After completing this brick, you will be able to:

1. Explain the mechanism of action of Angiotensin Converting Enzyme Inhibitors (ACE-Is) and Angiotensin Receptor Blockers (ARBs).
2. Compare and contrast pharmacologic profiles of ACE-Is and ARBs.
3. Discuss the clinical indications of ACE-Is and ARBs.
4. Outline the management strategies associated with common adverse effects of ACE-Is and ARBs.
5. Identify and explain the drug interactions and contraindications associated with the use of ACE-Is and ARBs.

CASE CONNECTION

EW is a 56-year-old male presenting to your urgent care clinic. He complains of a "dry, hacking cough" that has been going on for about eight weeks now. Initially, EW thought he "caught something" from his granddaughter when the cough started. But, the cough has not gone away and he is fed up. Of note, EW was also diagnosed with stage 1 hypertension 12 weeks ago, and his PCP prescribed lisinopril 10 mg daily for treatment. He is hoping you can prescribe something so that he can get rid of the cough!

Do you think that EW has a respiratory infection, or is something else causing his cough? How would you explain to EW why he has a cough? Consider your answers as you read, and we'll revisit EW at the end of the brick.

GO TO CONCLUSION ↓

Before we jump in to ACE inhibitors and ARBs, let's briefly review the RAAS.

What is RAAS?

The renin-angiotensi[illegible] interlinked hormonal pathway that is critical in maintaining our blood volume and blood pressure (BP). It affects several determinants of BP, such as intravascular volume, total body sodium, and blood vessel tone (constriction). The RAAS primarily determines serum levels of aldosterone and angiotensin II, two hormones that are key in regulating BP and renal blood flow (RBF). The **RAAS pathway** is a stepwise series of biochemical reactions that leads to formation of two main hormones, **angiotensin II** and **aldosterone** (Figure 1).

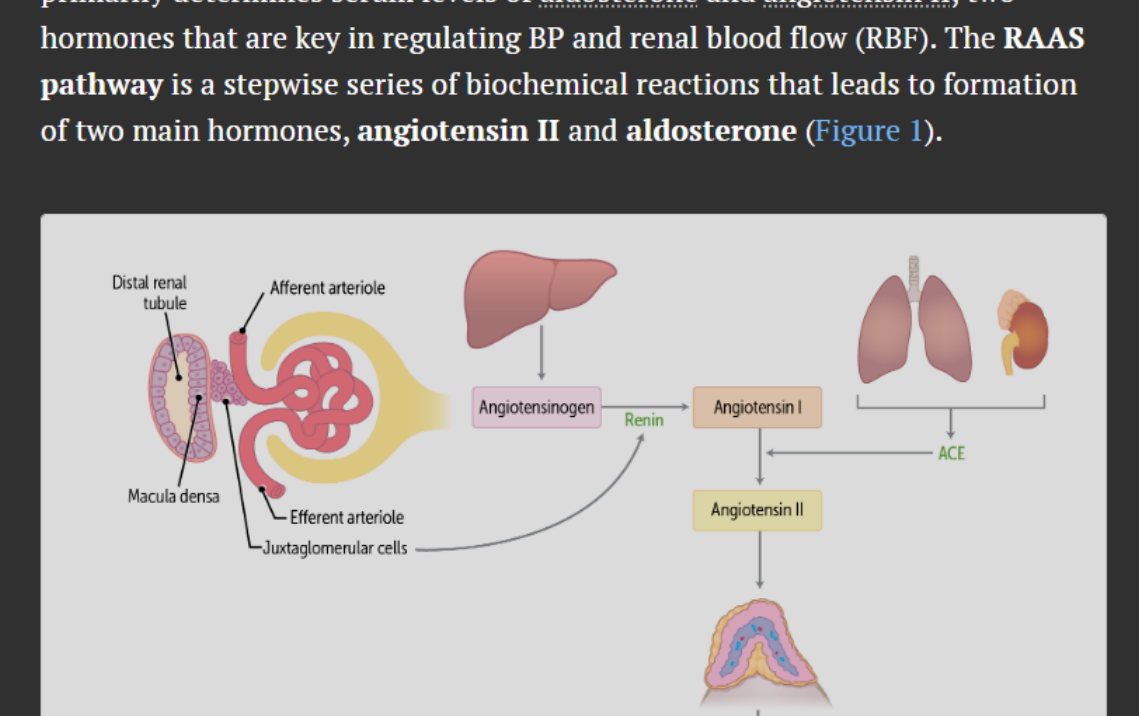


Figure 1

Let's take a brief look at the steps of the RAAS pathway.

1. **Release of renin.** Secretion of the protease enzyme renin by the kidney's juxtaglomerular (JG) cells gets things started. The JG cells, which are modified smooth [illegible] eriole, sense the **low blood flow, blood volume, or BP** that cause renin release.
2. **Formation of angiotensin I.** Once released, renin travels to the liver, which makes the inactive protein angiotensinogen. In the liver, renin cleaves angiotensinogen into angiotensin I, a precursor hormone to angiotensin II (Ang II).
3. **Formation of angiotensin II.** Angiotensin I travels to the lungs, where **angiotensin-converting enzyme (ACE)** converts angiotensin I into the peptide hormone **Ang II.**
4. **Release of aldosterone.** Ang II stimulates release of aldosterone from the adrenal gland. Aldosterone has an important role in expanding blood volume.

Both ***ACE and Ang II*** are important drug targets that will be discussed further in this brick.

How do ACE–Is and ARBs work? (LO#1)

Activation of the RAAS increases both blood pressure and total body sodium (and therefore extracellular and blood volume). Drugs that inhibit distinct portions of the RAAS include **angiotensin-converting enzyme (ACE) inhibitors, angiotensin-receptor blockers (ARBs),** and renin inhibitors. ACE inhibitors, ARBs, and renin inhibitors block these effects at various sites along the pathway, making them effective at lowering blood pressure (Figure 2). *Renin inhibitors are mentioned here for context, but, given their low use in clinical practice and only one agent (aliskiren) being available in the US, the remainder of the brick will focus on ACE inhibitors and ARBs.*

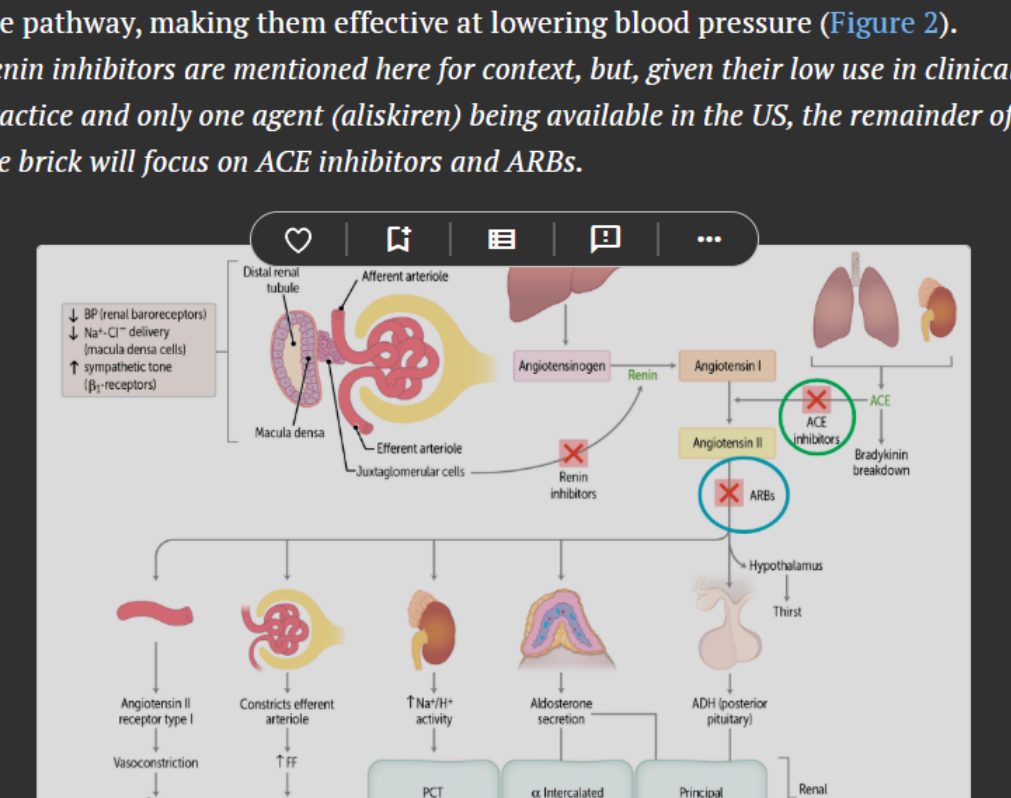

↓ BP (renal baroreceptors)
↓ Na⁺-Cl⁻ delivery (macula densa cells)
↑ sympathetic tone (β₁-receptors)
Distal renal tubule
Afferent arteriole
Macula densa
Efferent arteriole
Juxtaglomerular cells
Angiotensinogen
Renin
Renin inhibitors
Angiotensin I
ACE
ACE inhibitors
Bradykinin breakdown
Angiotensin II
ARBs
Hypothalamus
Thirst
Angiotensin II receptor type I
Vasoconstriction
↑ BP
Constricts efferent arteriole
↑ FF
Preserves GFR (when RBF ↓)
↑ Na⁺/H⁺ activity
PCT cell
Na⁺, HCO₃⁻, and H₂O reabsorption (permits contraction alkalosis)
Aldosterone secretion
α Intercalated cell
H⁺ secretion (↑ H⁺ ATPase activity)
ADH (posterior pituitary)
Principal cell
Na⁺ reabsorption K⁺ secretion (↑ K⁺ conductance, Na⁺/K⁺ ATPase, and ENaC activity)
H₂O reabsorption (via aquaporins)
Renal cells

Figure 2

What are the main drug classes that inhibit the RAAS pathway?

What are the pharmacologic differences between ACE–Is and ARBs? (LO#2)

As discussed above, ACE inhibitors and ARBs have the same downstream effects of lowering blood pressure. However, there are some slight pharmacologic differences between these two classes that are important to keep in mind.

ACE Inhibitors. ACE inhibitors target angiotensin-converting enzyme (ACE), the enzyme that catalyzes the conversion of angiotensin I to Ang II. As shown in Figure 2 above. The key downstream effects of ACE inhibitors include:

- Ang II receptors are located throughout the vasculature, and, under normal circumstances, they cause vasoconstriction and increase blood pressure. By reducing the formation of Ang II with ACE inhibitors, blood vessels are allowed to dilate and this effectively reduces blood pressure.
- The efferent arteriole also contains a high concentration of Ang II receptors. Again, as ACE inhibitors block the formation of Ang II the efferent arteriole is dilated, this also aids in blood pressure reduction.
- Reducing production of Ang II and aldosterone also *decreases potassium* secretion. This can cause increases in serum potassium levels when a patient is taking ACE inhibitors.
- ACE plays a large role in the degradation of bradykinin, an inflammatory mediator (see far right side of Figure 2). The primary expression of ACE in the lungs and subsequent *bradykinin accumulation* from its inhibition results in some u[illegible]his will be discussed later in the brick.

ARBs. Ang II receptor blockers carry out their effects one step downstream of ACE inhibitors. Instead of inhibiting the formation of Ang II, they block its receptors, limiting its action.

- As mentioned above, Ang II receptors are located throughout the vasculature. Ang II is still produced in the RAAS pathway (Figure 2), however, the receptors are inhibited not allowing them to produce their vasoconstrictive effects. The net effect is a reduction in blood pressure.
- ARBs also have similar net effects to ACE-Is on the efferent arteriole and decreased potassium secretion.

CLINICAL REASONING

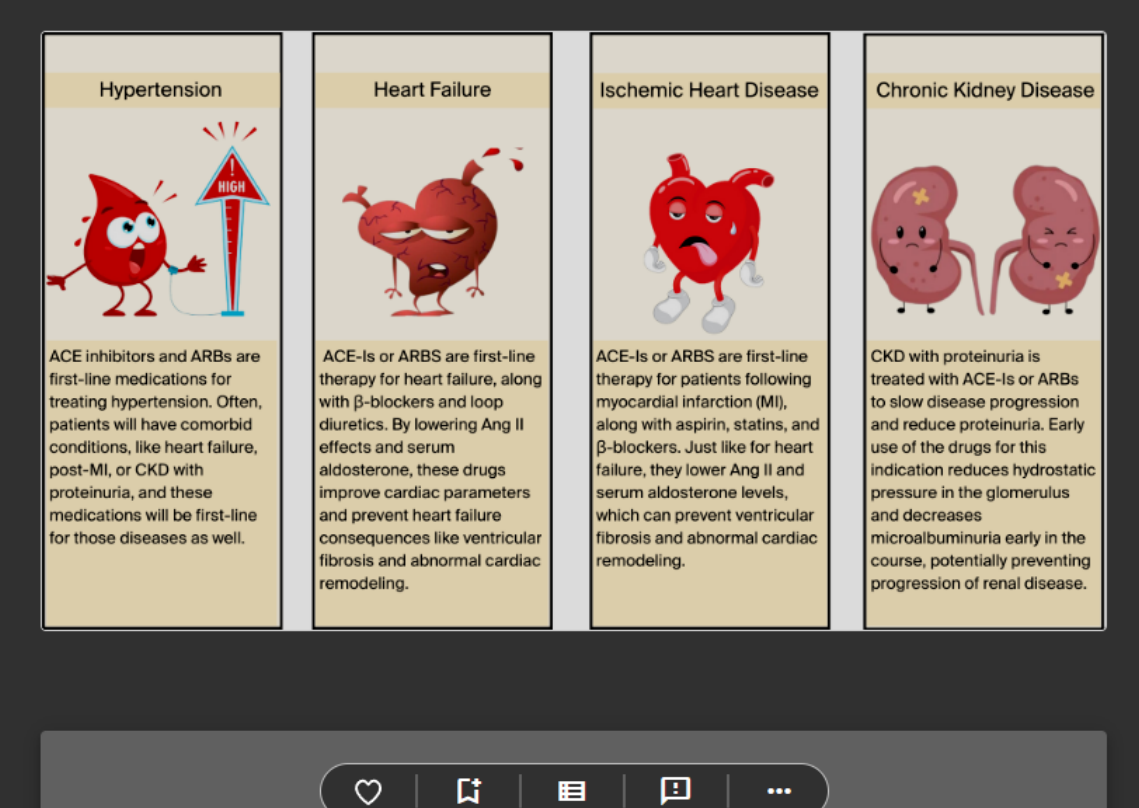
When would an ACE-I be preferred over an ARB? When would you choose an ARB over an ACE-I?

Both of these medication classes are effective at lowering blood pressure, and they should/ will be the most common ones you prescribe for that indication. However, note the slight pharmacologic difference that blocking the conversion of Angiotensin I to II (i.e., ACE-I) provides. This blocking of the conversion leads to an overall *decrease in the production of Ang II*, which is not the case with ARBs. The circulating levels of Ang II can still produce some vasoconstrictive effects on alternate angiotensin receptors. Clinically this means that ACE-Is tend to be more potent (i.e., will give a greater reduction in blood pressure), but they may have more side effects associated with them.

In general, ACE inhibitors and ARBs have an easily recognizable nomenclature. ACE inhibitors end with the suffix "-pril," and ARBs end with the suffix "-sartan." Table 1 lists commonly used drugs in each class.

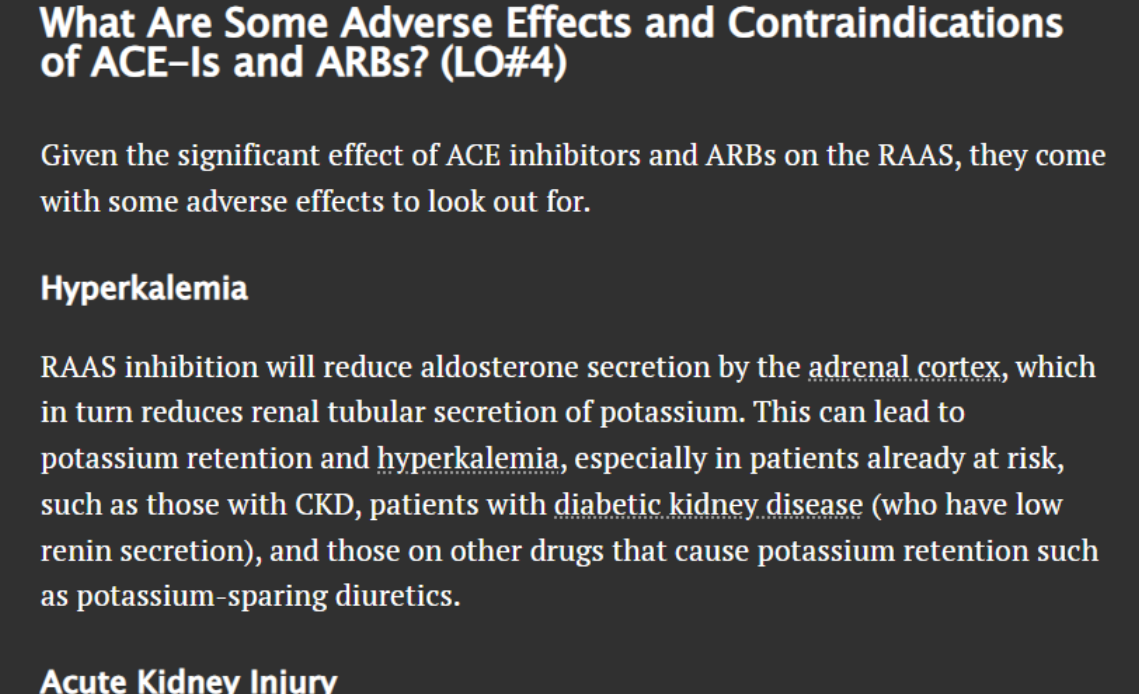
Table 1

ACE inhibitors	ARBs
Captopril	Valsartan
Enalapril	Losartan
Lisinopril	Candesartan
Benazepril	
Ramipril	


Place the ARBs and ACE inhibitor boxes in their appropriate positions on the graphic.

What Are the Indications for RAAS Inhibitors? (LO#3)

ACE inhibitors and ARBs are used most often for treatment of *hypertension*, but they are also used for patients with *chronic kidney disease (CKD) with proteinuria, post-myocardial infarction*, and for *heart failure*.



Hypertension

ACE inhibitors and ARBs are first-line medications for treating hypertension. Often, patients will have comorbid conditions, like heart failure, post-MI, or CKD with proteinuria, and these medications will be first-line for those diseases as well.

Heart Failure

ACE-Is or ARBs are first-line therapy for heart failure, along with β-blockers and loop diuretics. By lowering Ang II effects and serum aldosterone, these drugs improve cardiac parameters and prevent heart failure consequences like ventricular fibrosis and abnormal cardiac remodeling.

Ischemic Heart Disease

ACE-Is or ARBS are first-line therapy for patients following myocardial infarction (MI), along with aspirin, statins, and β-blockers. Just like for heart failure, they lower Ang II and serum aldosterone levels, which can prevent ventricular fibrosis and abnormal cardiac remodeling.

Chronic Kidney Disease

CKD with proteinuria is treated with ACE-Is or ARBs to slow disease progression and reduce proteinuria. Early use of the drugs for this indication reduces hydrostatic pressure in the glomerulus and decreases microalbuminuria early in the course, potentially preventing progression of renal disease.

Name 4 indications where ACE-Is/ ARBs are first-line therapy?

What Are Some Adverse Effects and Contraindications of ACE–Is and ARBs? (LO#4)

Given the significant effect of ACE inhibitors and ARBs on the RAAS, they come with some adverse effects to look out for.

Hyperkalemia

RAAS inhibition will reduce aldosterone secretion by the adrenal cortex, which in turn reduces renal tubular secretion of potassium. This can lead to potassium retention and hyperkalemia, especially in patients already at risk, such as those with CKD, patients with diabetic kidney disease (who have low renin secretion), and those on other drugs that cause potassium retention such as potassium-sparing diuretics.

Acute Kidney Injury

Under normal circumstances, angiotensin II works to increase the GFR via constriction of the efferent arteriole (Figure 3, left).

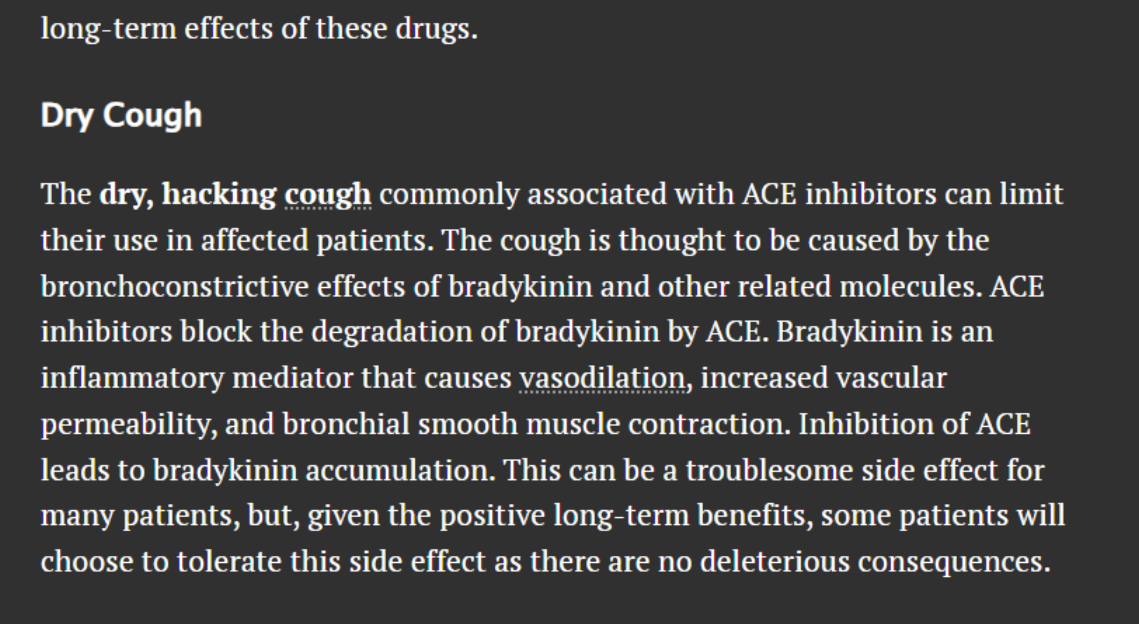

Normal Glomerular Filtration Rate
Vasoconstrictor Angiotensin II
Efferent arteriole
Afferent arteriole
Glomerular capillaries
Bowman capsule
Reduced Glomerular Filtration Rate
ACE inhibitors, ARBs, renin inhibitors
Vasoconstrictor Angiotensin II
Efferent arteriole
Afferent arteriole
Glomerular capillaries
Bowman capsule

Figure 3

Drugs targeting the RAAS will reduce angiotensin II levels or the effects, so they will dilate the efferent arteriole (Figure 3, right). This dilation will sometimes **acutely decrease GFR**, especially in low flow states like dehydration or heart failure. This is a form of **prerenal acute kidney injury (AKI).** As the name implies, prerenal refers to a cause occurring *before the kidney;* this most often occurs by reduced blood flow to the kidneys which reduces their ability to filter blood appropriately, and, as in the case with ACE-Is/ ARBs, may cause an AKI. The fall in GFR often occurs within the first few days of starting these medications. Most patients GFR will return to normal in 7-10 days, however, i[illegible]lose should be reduced or discontinued. Given the importance of these drugs in CKD and heart failure, an initial decrease in GFR may be acceptable given the overall positive long-term effects of these drugs.

Dry Cough

The **dry, hacking cough** commonly associated with ACE inhibitors can limit their use in affected patients. The cough is thought to be caused by the bronchoconstrictive effects of bradykinin and other related molecules. ACE inhibitors block the degradation of bradykinin by ACE. Bradykinin is an inflammatory mediator that causes vasodilation, increased vascular permeability, and bronchial smooth muscle contraction. Inhibition of ACE leads to bradykinin accumulation. This can be a troublesome side effect for many patients, but, given the positive long-term benefits, some patients will choose to tolerate this side effect as there are no deleterious consequences.

CLINICAL REASONING

Because of the role of ACE in the breakdown of bradykinin, ARBs have a reduced incidence of cough compared to ACE inhibitors. There are even some differences in the incidence of cough between the different ACE-Is. If a patient experiences a dry cough, then switching to a different ACE-I or ARB can alleviate this side effect. Also, this side effect can show up at any time, even years after they have been taking the medication.

Angioedema

A rare, but more serious complication of increased bradykinin is ACE inhibitor–associated **angioedema** ("angio-", meaning related to blood vessels, "-edema" meaning swelling). As mentioned above, ACE inhibitors block the degradation of bradykinin. This large buildup of bradykinin can bind to receptors in vascular endothelium, which stimulates substance P (neuropeptide that helps mediate inflammatory response in the body) release and induces angioedema.

This is a potentially serious side effect that commonly affects the upper airways and the face as well as the lips and tongue (Figure 4). Airway obstruction is the most feared complication of angioedema and can be fatal.

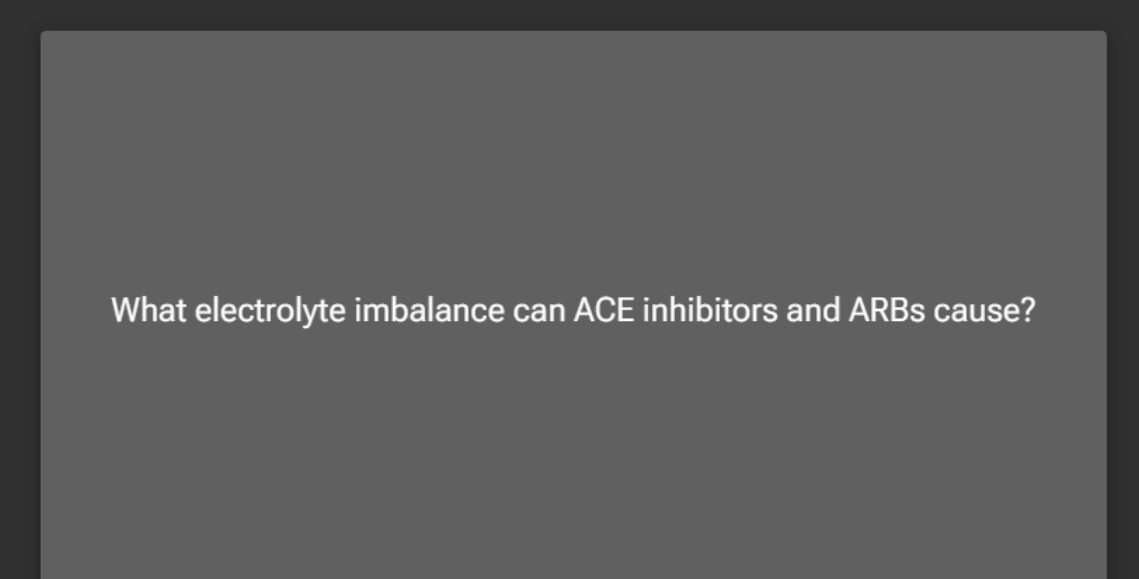


Figure 4

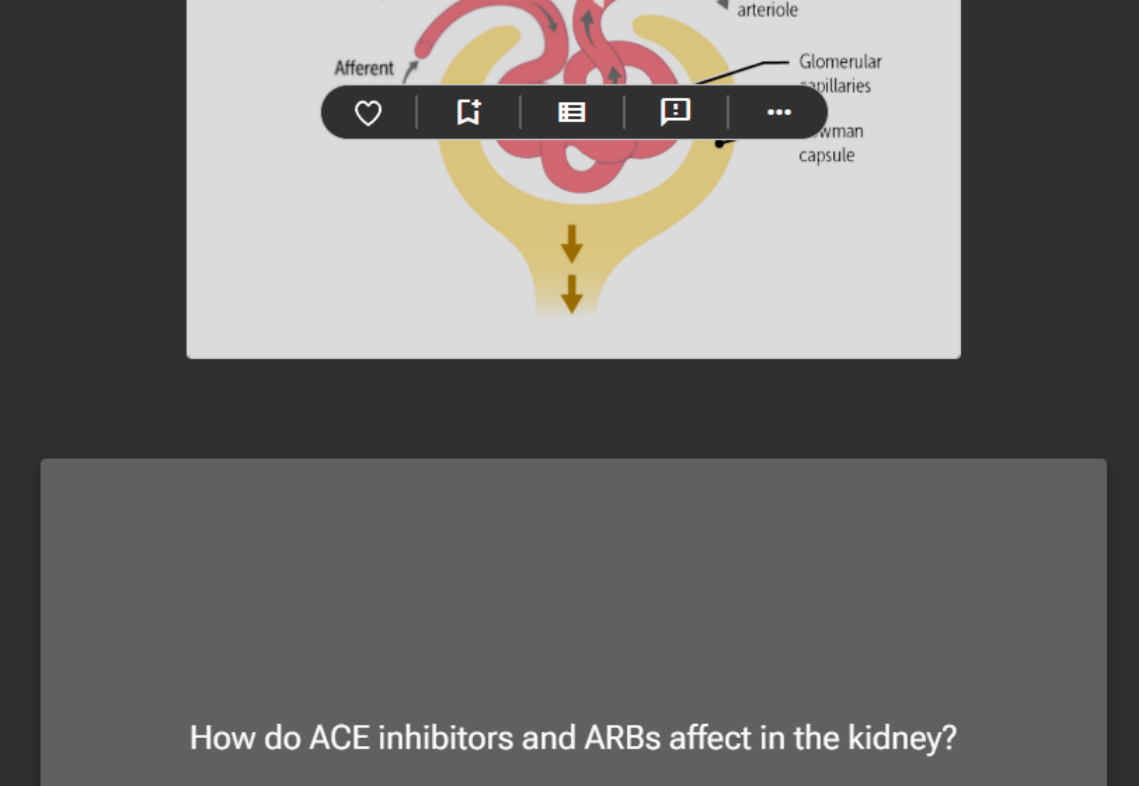
CLINICAL CORREL[illegible]

African American patients have a 4.5 times greater risk of ACEI-induced angioedema than Caucasians. If treating hypertension alone in an African American patient, generally, a DHP calcium channel blocker (e.g. amlodipine) will be the most preferred first-line agent. If comorbid conditions, like heart failure, are present with the hypertension, then a trial with an ARB, and close monitoring, is generally warranted.

What electrolyte imbalance can ACE inhibitors and ARBs cause?

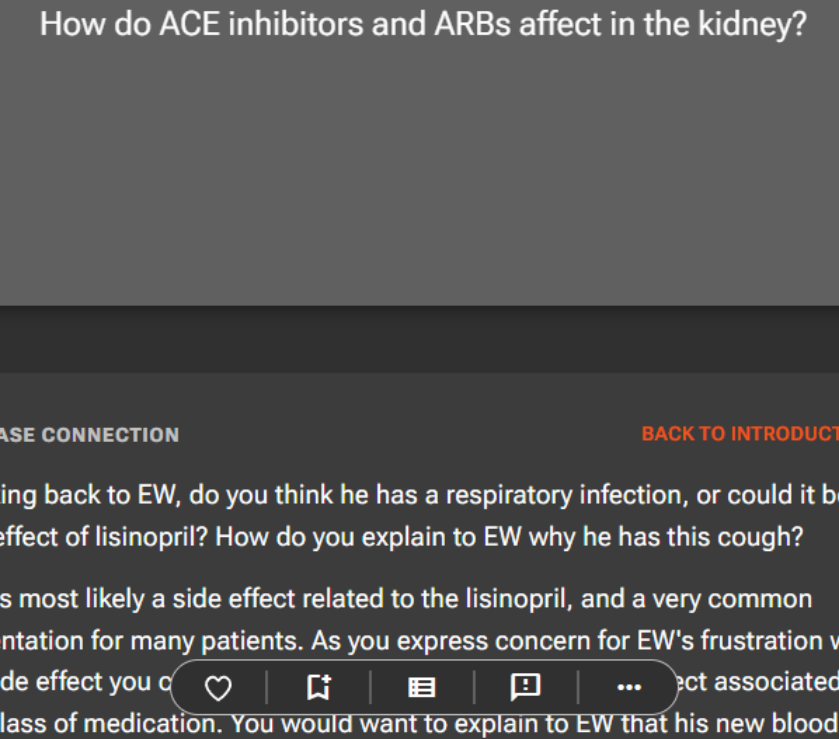
What are the contraindications and drug interactions of ACE–Is and ARBs? (LO#5)

ACE-Is and ARBs **should NOT be used together** because of additive side effects like hyperkalemia and AKI, as well as an increased risk for angioedema. Additionally, blockin[illegible] been shown to have any additional clinical benefit.



ACE-Is & ARBs

Contraindications	Drug Interactions
Pregnancy • Contraindicated in pregnancy due to their teratogenicity • They can cause fetal kidney injury	**Hyperkalemia** • On their own, ACE-Is and ARBs can cause hyperkalemia, but this can be worsened when they are taken with other agents, like: • Aldosterone antagonists (spironolactone) • Sodium channel blockers (amiloride and triamterene) • Potassium Supplements
Bilateral Renal Artery Stenosis • **Contraindicated in bilateral renal artery stenosis** due to the low renal blood flow and risk for further declining renal function.	**Acute Kidney Injury** • Taking ACE-Is / ARBs along with NSAIDs can lead to AKI. NSAIDs inhibit the synthesis of prostaglandins, which normally dilate the afferent arteriole. Blockage of this afferent arteriolar dilation coupled with the dilation of the efferent arteriole by ACE inhibitors reduces GFR, especially in patients who are already at risk for kidney injury.


Reduced Glomerular Filtration Rate
ACE inhibitors, ARBs, renin inhibitors
Vasodilator PGs (PGE₂ PGI₂)
NSAIDs
Vasoconstrictor Angiotensin II
Efferent arteriole
Afferent
Glomerular capillaries
Bowman capsule

How do ACE inhibitors and ARBs affect in the kidney?

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to EW, do you think he has a respiratory infection, or could it be a side effect of lisinopril? How do you explain to EW why he has this cough?

This is most likely a side effect related to the lisinopril, and a very common presentation for many patients. As you express concern for EW's frustration with the side effect you c[illegible]ect associated with this class of medication. You would want to explain to EW that his new blood pressure medication, lisinopril, lowers blood pressure by blocking the substance ACE (hence the medication class ACE inhibitor) that is key in formation of another substance that usually constricts the blood vessels. By blocking ACE it allows the blood vessels to dilate, and that lowers blood pressure. However, ACE normally breaks down another inflammatory substance called bradykinin. If ACE is blocked, then a buildup of bradykinin can irritate the airways and cause the dry cough EW is experiencing.

If it is something that bothers him, and, possibly, affects his perceived quality of life then we could consider switching to a different medication within the class or try an ARB with a lower incidence of the cough.

Summary

- The renin-angiotensin-aldosterone system (RAAS) is a hormonal pathway that plays a crucial role in maintaining blood volume and blood pressure. It primarily determines serum levels of aldosterone and angiotensin II.
- Drugs that inhibit the RAAS include angiotensin-converting enzyme (ACE) inhibitors and angiotensin-receptor blockers (ARBs). These drugs are effective at lowering blood pressure.
- ACE inhibitors block the angiotensin converting enzyme with catalyzes the conversion of angiotensin I to Ang II. The reduction in Ang II leads to a dilation of the efferent arteriole, which reduces blood pressure.
- ARBs block Ang II receptors, which, again, dilate the efferent arteriole and reduce blood pressure.
- ACE inhibitors and ARBs are commonly used for treating hypertension, chronic kidney d[illegible]ial infarction, and heart failure.
- Common side effects of ACE inhibitors and ARBs include hyperkalemia, acute kidney injury, dry cough, and angioedema. ARBs have a lower risk for dry cough and angioedema compared to ACE inhibitors.
- ACE inhibitors and ARBs should not be combined due to the risk of additive side effects such as renal injury. They are contraindicated in bilateral renal artery stenosis and in pregnant women.
- ACE inhibitors and ARBs should not be used alongside potassium-sparing diuretics, oral potassium supplements, or chronic NSAIDs due to the risk of hyperkalemia and prerenal acute kidney injury.

Review Questions

1. Which drug class inhibits the conversion of angiotensin I to angiotensin II?

A. ● ACE Inhibitors
B. ○ ARBs
C. ○ Renin inhibitors
D. ○ Diuretics
E. ○ Beta-blockers

Hide Explanation

The correct answer is ACE inhibitors (A). ACE inhibitors block the enzyme responsible for converting angiotensin I to angiotensin II, thereby lowering blood pressure. ARBs (B) block the angiotensin II receptor. Renin inhibitors (C) block the activity of renin. Diuretics (D) increase urine output to reduce blood volume. Beta-blockers (E) reduce heart rate and cardiac output.

2. Which effect is caused by ARBs?

A. ● Dilation of blood vessels
B. ○ Increased vasoconstriction
C. ○ Increased blood pressure
D. ○ Increased potassium secretion
E. ○ Bradykinin degradation

Hide Explanation

The correct answer is dilation of blood vessels (A). ARBs reduce the formation of Ang II, leading to vasodilation and reduced blood pressure. Increased vasoconstriction (B) is incorrect as ARBs prevent this. Increased blood pressure (C) is incorrect because ARBs lower blood pressure. Increased potassium secretion (D) is incorrect as ARBs decrease potassium secretion. Bradykinin degradation (E) is incorrect as ACE is involved in this process.

3. Which medication class is used to reduce proteinuria in CKD?

A. ● ACE-Is or ARBs
B. ○ Diuretics
C. ○ Beta-blockers
D. ○ Calcium channel blockers
E. ○ Statins

Hide Explanation

The correct answer is ACE-Is or ARBs (A). These medications reduce hydrostatic pressure in the glomerulus and decrease microalbuminuria, slowing disease progression. Diuretics (B) are used to manage fluid balance but do not specifically reduce proteinuria. Beta-blockers (C) are primarily used for cardiovascular conditions. Calcium channel blockers (D) manage hypertension but are not first-line for proteinuria. Statins (E) lower cholesterol but do not directly reduce proteinuria.

4. What causes the dry cough associated with ACE inhibitors?

A. ○ Reduced angiotensin II
B. ● Bradykini[illegible]
C. ○ Increased GFR
D. ○ Efferent arteriole dilation
E. ○ Increased vascular permeability

Hide Explanation

The correct answer is bradykinin accumulation (B). ACE inhibitors block the degradation of bradykinin, leading to its accumulation and causing a dry cough. Reduced angiotensin II (A) and increased GFR (C) are not directly related to the cough. Efferent arteriole dilation (D) and increased vascular permeability (E) are effects of ACE inhibitors but do not cause the cough.

5. Which medication class should generally be avoided when taking ACE-Is/ ARBs?

A. ○ Beta-blockers
B. ○ Calcium channel blockers
C. ○ Loop diuretics
D. ● NSAIDs

Hide Explanation

The correct answer is NSAIDs (D). Recall that ACE-Is/ ARBs dilate the efferent arteriole, while NSAIDs constrict the afferent arteriole. This overall reduction in renal blood flow when these medications are combined greatly increase the risk for AKI and the combination should be avoided. Beta-blockers (A), Calcium channel blockers (B), and Loop diuretics (C) are all commonly used in combination with ACE-Is/ ARBs and do not produce any notable drug interactions.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Explain the mechanism of action of Angiotensin Converting Enzyme Inhibitors (ACE-Is) and Angiotensin Receptor Blockers (ARBs).~~
02 ☑ ~~Compare and contrast pharmacologic profiles of ACE-Is and ARBs.~~
03 ☑ ~~Discuss the clinical indications of ACE-Is and ARBs.~~
04 ☑ ~~Outline the management strategies associated with common advers~~

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Explain the mechanism of action of Angiotensin Converting Enzyme Inhibitors (ACE-Is) and Angiotensin Receptor Blockers (ARBs).~~
02 ☑ ~~Compare and contrast pharmacologic profiles of ACE-Is and ARBs.~~
03 ☑ ~~Discuss the clinical indications of ACE-Is and ARBs.~~
04 ☑ ~~Outline the management strategies associated with common adverse effects of ACE-Is and ARBs.~~
05 ☑ ~~Identify and explain the drug interactions and contraindications associated with the use of ACE-Is and ARBs.~~

Objective Confidence: 5/5